

MAZAK GEAR BOX REPLACEMENT

Requirements Document

For

704th Test Group/846th Test Squadron

11 December 2024

1.0 PURPOSE

The 704th Test Group's (704 TG) mission is to be the premier provider of unique test and evaluation capabilities relevant to tomorrow's warfighter through world-class people improving technical excellence, cost-effectiveness, and agility. The 846th Test Squadron (846 TS) is one of the units assigned as part of the 704 TG. The 846 TS's primary test capability is the Holloman High Speed Test Track (HHSTT) where they utilize rocket powered sled trains to simulate flight like environments and scenarios with the control and recovery of a ground-based system.

The 846 TS uses a Mazak VCN-510C50-11 CNC Mill, serial number 199570, to machine complex rocket sleds, test track hardware and test articles. Currently, the machine's main milling spindle gear box have become worn and inoperative and will no longer engage any spindle or tool change.

The 846 TS is requesting a re-manufactured gear box from the original equipment manufacturer (OEM) "The Mazak Corporation" as the replacement part for this repair. A new replacement gear box is unavailable, and the re-manufactured item is the best option to return the machine back to factory specification.

2.0 SCOPE

The contractor shall meet the following requirements:

- 2.1 Provide an authorized MAZAK technician to remove the damaged gear box (MAZAK Part # 556501R19C0) from the Mazak CNC Mill, model number VCN-510C50-11, serial number 199570 and replace it with an OEM re-manufactured gear box, on-site at Holloman AFB.
- 2.2 The re-manufactured gear box must come with a 1-year warranty.
- 2.3 Perform acceptance runoff: An authorized Mazak technician will perform machine and spindle operation after installation by running spindle for 30 minutes continuous, make test cuts to ensure accuracy, and perform proper tool change operations.

3.0 MILESTONES

The contractor shall complete installation of OEM re-manufactured gear box and acceptance runoff no later than 45 days ARO.

4.0 DELIVERABLES

The contractor shall deliver all items to 846 Test Squadron, Attn: J. Pieszala at 1521 Test Track Road, Building 1173, Holloman Air Force Base, New Mexico 88310.